On the Robustness of Physics-Informed Neural Operators: Super Resolution and Noisy Data Regimes

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Abstract

The physics-informed neural operator (PINO) is a deep learning method for solving partial differential equations (PDE). PINO incorporates a hybrid loss function that combines both data and physics constraints to learn an operator that maps initial conditions to the solution function. PINO is claimed to accurately predict beyond the resolution of its training data and succeed in certain low-data regimes because it captures dynamic laws of the system. We test the validity of this claim under zero-shot super resolution, as well as provide a justification for monotone convergence in the limit of mesh refinement. We find that zero-shot super resolution holds robustly up to 32x the training resolution, but breaks down for very coarse training meshes ($N_x \leq 16$) where data conflicts with the physics residual rather than complementing it. Traditionally, the PINO framework has been used for engineering problems with high data fidelity. We assess its capabilities under rough data through the Black-Scholes equation, using historical option pricing data. Under the same number of training samples, real-world market data produces solution profiles broadly comparable to analytic data, though whether it can serve as a reliable substitute for high-fidelity supervision remains an open question. Both of these results provide strong indicators of the robustness of this architecture, and serve to justify its use in new, non-ideal conditions. Our code is available at `https://github.com/hishambhatti/PINO-apps`.

1 Introduction

Physics-informed machine learning has emerged as a paradigm to science and engineering problems governed by partial differential equations (PDEs) (Hao et al., 2022). Traditional numerical methods such as finite difference and finite element discretize the sample space into a mesh of points or shapes, and iteratively propagate the solution function. As such, these methods are often slow, with computation time growing exponentially with dimension. Moreover, discretization-dependence creates a tradeoff between accuracy and speed as the mesh resolution of grid points changes.

Physics-informed neural networks (PINN) offer a breakthrough for this problem by using deep neural networks to learn a continuous solution function that respects physical laws of the system (Raissi et al., 2019). However, standard neural networks are often data-hungry, which can be problematic in scientific problems where data is costly, time-consuming, or potentially dangerous to obtain, and is therefore scarce. Along with standard MSE from human-collected data, physics-informed neural networks also directly embed physics into their loss function. Auto-differentiation can be used to obtain derivatives of output values with respect to input coordinates. These values can be directly plugged into the known PDE to quantify the deviation from satisfying the law known as the physics residual, forming a physics component of loss. This term acts as regularization, biasing the model away from the data to respect the known physical laws.

Physics-informed neural networks have shown to be orders of magnitude faster at inference time compared to standard numerical methods. The lack of a mesh renders this solution suitable for

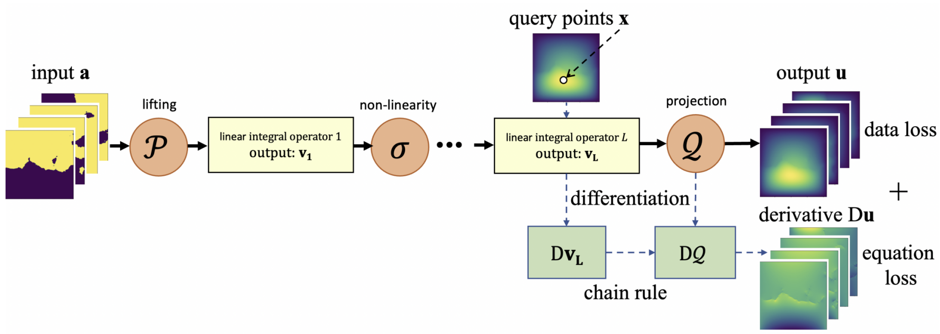


Figure 1: PINO trains neural operator with both training data and a PDE loss function. Both v_L and u are functions and all its derivatives (Dv_L, Du) can be computed in exact form.

high-dimensional problems. Since physics loss can be computed at any sample point, these methods have shown promise in the small data regime, where human-collected data is coarse or nonexistent. However, in practice, the optimization landscape is tricky to converge (Krishnapriyan et al., 2021). PINNs are often sensitive to hyperparameters and initializations, where traditional grid methods can potentially be faster. The solution function obtained through a PINN is also tied to a specific instance of initial conditions, so that adjusting the system slightly involves retraining the network from scratch.

A recent breakthrough is the use of operator learning for solving PDEs (Kovachki et al., 2023). Unlike neural networks which learn a function mapping inputs and outputs of fixed dimensions, neural operators learn operators, which are mappings between function spaces. These operators are a more natural object when studying PDE solutions. For a PDE, the solution operator is the mapping from the initial function (initial and boundary conditions) to the output solution function. This framework allows us to change the conditions of our system without the need for retraining. Additionally, neural operators are proven to be discretization convergent in the limit of mesh refinement, meaning they converge to a continuum operator as the discretization is refined.

Widely used is the Fourier Neural Operator (FNO) (Li et al., 2021). Rooted in spectral solvers, which decompose a solution into sinusoidal functions, this architecture uses the fast Fourier transform (FFT) to convert the input into frequency space, perform linear operators in that space, then inverse Fourier transform and output the solution function. The Fourier transform decomposes the initial function into its component wavefunctions, and the FNO assigns trainable weights to each of these wavenumbers. In practice, all higher-order wavenumbers above a threshold ($k = 12$) are ignored as noise, which acts as another form of regularization. The benefit of this architecture is two-fold. One, the FFT runs in quasi-linear time, providing similar speedups during inference time compared to numerical solvers. Two, the Fourier modes learned represent global patterns in the solution function, a more natural approach for understanding a PDE. Additionally, these modes capture global oscillations and are *resolution-invariant*. This crucial property allows the same model to be evaluated, trained, or inferred across different spatial resolutions. These methods have found applications in a wide variety of scientific problems, particularly those with global patterns such as weather and climate (Choi et al., 2024; Bonev et al., 2023).

However, the fundamental approach is still data-driven, with a lack of generalization outside of the data regime. The Physics-Informed Neural Operator (PINO) provides a framework to remedy this shortcoming. As shown in Figure 1, it constructs a hybrid loss function combining data loss from the neural operator with the physics residual from the physics-informed network (Li et al., 2022). In doing so, it provides the same benefits of generalization from PINNs, along with the resolution-invariance, stable optimization, and learned family of solutions of the Fourier Neural Operator. This approach has been shown to approximate the solution operator of many PDE families nearly perfectly, and has been used in a variety of applications (Rosofsky et al., 2023).

In this paper, we stress-test the robustness of the PINO framework under extreme and non-ideal data regimes to better understand what kinds of problems it is capable of solving. Specifically, our contributions are two-fold:

- **Zero-Shot Super-Resolution:** We evaluate the accuracy of PINO at drastically increasing test resolutions to identify the practical breaking points of its discretization transfer capabilities.
- **Real-World Empirical Data Regimes:** We assess PINO's performance when regularized by a noisy, empirical economics framework loosely guided by the Black-Scholes equation.

2 Capabilities of the PDE Loss: Zero-Shot Super Resolution

Since real-world data from physical systems can be costly to obtain, training data is often sparse, consisting of measurements of interest on a coarse grid. However, we want our systems to be representable as continuous functions, so that if we query our solutions at arbitrary points in between points of the training mesh, accuracy is maintained. This is the core idea of *zero-shot super resolution*: train with data at a fixed coarse resolution (constrained by real-world data limitations), but sample, compute, and backpropagate PDE loss on a higher resolution grid. This way, the accuracy does not degrade when evaluating an inference mesh with a higher resolution than the training data but at or below the resolution of the PDE grid.

Part of the FNO architecture involves learnable weights for a set limited number of wavelengths, which correspond to the number of oscillations of the solution function over the system mesh. Higher order modes were thrown away as a form of regularization. In our study, for an N_x point grid, the number of modes was manually set to $N_x/2$, so that the FNO only learns weights for the lowest mode wavenumbers. With this framework, zero-shot super resolution asserts that high resolution PDE samples guide the fixed Fourier modes to be physically consistent across the domain. For example, if you train with training data $N_x = 128$ and PDE loss at higher resolution, the model learns to be consistent with physics in those $N_x/2 = 64$ modes when evaluated on a finer grid, and accuracy doesn't degrade. As a note, training at a higher PDE resolution does *not* suggest that our model will learn higher-frequency physical behavior that was not captured in coarse training data. It is known that the Fourier neural operator cannot perfectly approximate the ground truth operator when only coarse training data is provided (Li et al., 2021). Instead, a higher physics resolution serves to propagate the function through the mesh as a better way than interpolation, so that our error stays consistent across inference meshes. Therefore, in our experiments probing zero-shot super resolution, we artificially cap the Fourier wavelengths to that of the training data $N_x/2$ when performing these tests. This was implicitly done in the original paper (Li et al., 2022). Doing so allows us to answer questions about whether the model satisfies the wave equation in the modes it was trained on, not whether it produces physically correct content in higher-order modes at super-resolution.

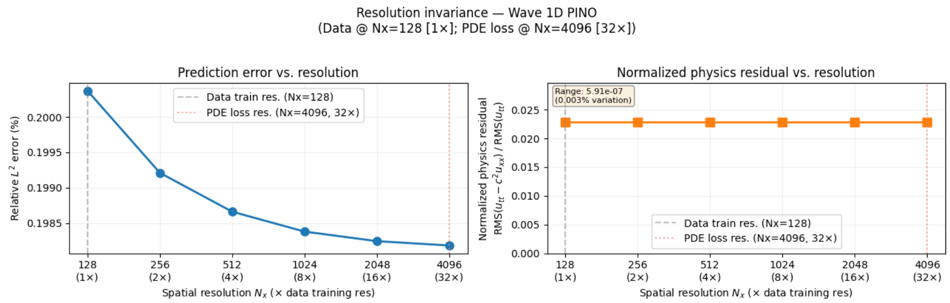
With this framework, we aim to effectively compare the performance of the PINO to solve the 1D Wave Equation across an exponentially increasing set of resolutions. Specifically, we aimed to answer the following questions:

- Does zero-shot super resolution still hold up to 32x the training data resolution $N_x = 128$?
- How does performance compare when trained at a fixed data and PDE resolution, and evaluated on increasing resolutions?
- Why does convergence overwhelmingly happen monotonically as resolution increases?
- How coarse can the training data be such that zero-shot resolution still holds up to 16x?

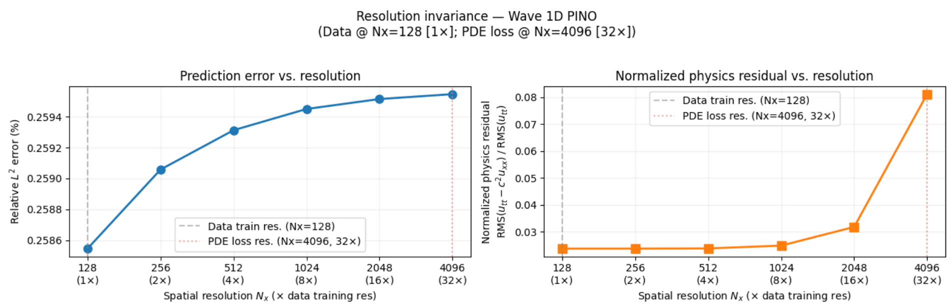
2.1 Extension of Zero-Shot Super Resolution

The original paper claimed zero-shot super resolution, but only evaluated the claim on a problem involving Kolmogorov flows up to 4x the training resolution. In practice for expensive systems, training data can be orders of magnitude coarser than our intended predictions. Additionally, the Kolmogorov Flow problem is inherently a coupled system, so that low-frequency Fourier modes in our architecture may inform higher wavelengths, thereby corrupting the results of zero-shot super resolution with capped Fourier modes.

To address these challenges, we tested zero-shot super resolution of the linear, decoupled 1D Wave Equation up to 32x the training resolution (Figure 2a). The training data was fixed at $N_x = 128$ points. As expected, the physics residual remained consistent as we trained at a higher resolution



(a) Capped wavenumbers



(b) Uncapped wavenumbers

Figure 2: Zero-shot super resolution test for 1D wave equation

and just sampled fewer points at lower resolutions (deviations are on order 1e-7, which we attribute to experimental noise). Notably, the L^2 error converged rather than exploded, with differences on orders of 1e-4, suggesting that zero-shot super resolution holds for this system.

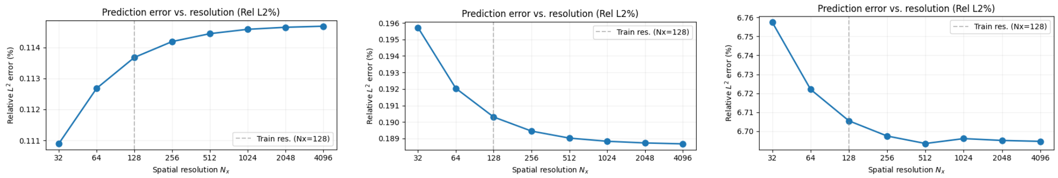
We also ran an ablation study where wavelengths were not capped to $k = 64$ at increasing resolutions (Figure 2b). At higher resolution, physics residual loss grows exponentially, suggesting that the model does not generate physically correct content for modes above 64. Little deviations in higher modes (like due to floating point rounding errors), get multiplied by k^2 when taking second-order spectral derivatives. While physics residual for higher modes may be negligible in the predicted solution, at $N_x = 4096$ (32x resolution), a quadratically growing noise term creates non-negligible physics residual. This is not an indication of zero-shot super resolution failing, but rather a fundamental limitation of fixing the number of Fourier modes.

2.2 Zero-Shot Discretization Transfer at High Resolutions

We also aim to understand whether the operator learned at coarse training and PDE resolution retains fidelity when queried at resolutions far beyond the training distribution, without any fine-tuning of physics loss at higher resolution. Note that this is *not* a test of zero-shot super resolution. We don't train with PDE residual any finer than the training data, but simply evaluate performance when the same model is tested and averaged across more points. For this experiment, we trained PINO on data with spatial resolution $N_x = 128$, and evaluated the model on data with progressively higher spatial resolution, ranging from $N_x = 32$ up to $N_x = 4096$. The results of this experiment are shown in Figure 3. The relative L^2 error was similarly low at every resolution we evaluated, varying on orders of 1e-2 or lower percentages points in each trial.

2.2.1 Analysis of Monotonic Convergence in High Resolutions

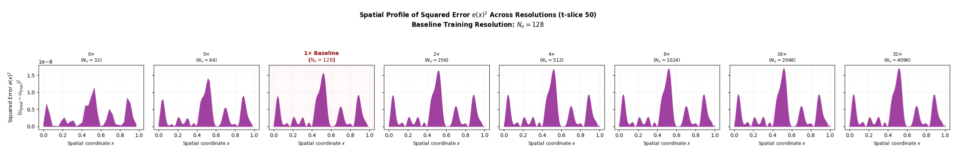
In these previous experiments, we noticed an interesting trend: all evaluations converged, but the vast majority of runs converged monotonically either upwards or downwards (Figure 3a, 3b). Convergence of the L^2 error is expected, as the result of two separate convergences: model output and ground truth. The model output converges to a continuous function because the FNO architecture is



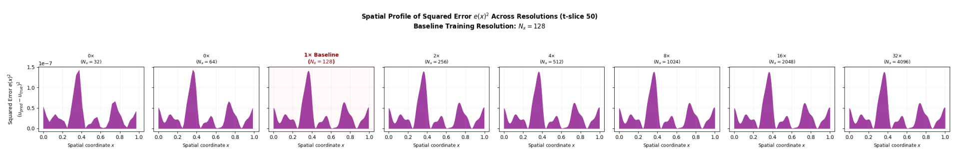
(a) Upward convergence (b) Downward convergence (c) Non-monotone convergence

Figure 3: Convergence trends for models trained at fixed training resolution and PDE resolution ($N_x = 128$) when evaluated at higher resolutions.

convergent in the limit of mesh refinement (Li et al., 2021). As we change the sample grid of input points, the model receives a finer encoding of the initial condition. The FNO's spectral convolutions operate on a more global representation, so that the output function itself changes. At the same time, ground truth converges to the continuous, analytical solution function. The L^2 error function, corresponding to the squared different between them, converges accordingly. Intuition might suggest that the error functions will be noisy, a mixture of convex and concave regions, leading to messy convergence as shown in 3c. But in practice, convergence overwhelmingly occurs uni-directionally.



(a) Squared error in upward convergence



(b) Squared error in downward convergence

(c) Squared error in non-monotone convergence

Figure 4: The squared error function for different convergences in models trained at fixed training resolution and PDE resolution ($N_x = 128$) when evaluated at higher resolutions.

Whether error converges upward, downward, or neither depends on the relative speed of model and ground truth convergence and in which direction, determined by the behavior of the largest peaks in the error landscape. This hypothesis was supported in Figure 4. The error landscape for monotone convergence features one dominant peak (Figure 4a, 4b), whereas the non-monotone convergence has multiple peaks of similar height and width (Figure 4c). For errors with one peaked function, convergence is guided by the peak. If the peak grows or widens, we observe upward convergence; if the peak shrinks, narrows, splits, we get downward convergence. Non-monotone convergence only occurs when peaks are comparable in height, creating conflicting regions of upward and downward convergence. This occurs rarely in practice. In our experiments, such a scenario was only reproducible under highly unstable optimization trajectories characterized by severe numerical instability, where the raw residual errors spanned orders of magnitude higher than those of stable training runs.

Since the peaked landscape tends to persist across resolutions, the majority of training runs converge monotonically, guided by the change in area of the largest error peak.

2.3 ASSESSMENT OF ZERO-SHOT SUPER RESOLUTION TO COARSE TRAINING DATA

Given that zero-shot super resolution is achieved to at least 32x the training resolution of $N_x = 128$, a follow-up question asks when this break down. Specifically, as we decrease the number of training data points, how does zero-shot super resolution behave. Such a question helps us understand the role of data samples in the PINO architecture, and how this translates to low-data regimes.

To address this, we trained models with training resolution from $N_x = 4$ to $N_x = 128$, and PDE loss at 16x the respective training loss. Each model was evaluated at a test resolution grid up to 16x that of the training data. The goal was to understand how coarse training data can be to exhibit this property. We also trained and evaluated a baseline model with no data and just PDE loss at 16x the highest training resolution of $N_x = 128$, to assess whether data itself is needed in the pipeline.

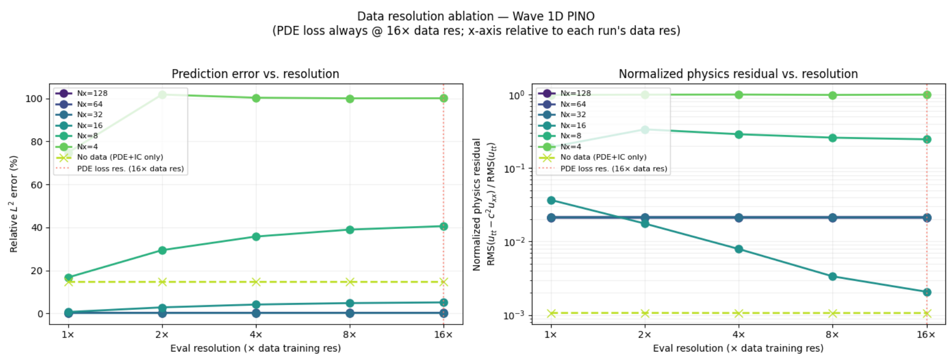


Figure 5: An assessment of zero-shot super resolution for various training data resolutions N_x

With mode capping and normalization, we obtain the results shown in Figure 5. The no data model has the lowest PDE loss, roughly an order of magnitude lower than other models. This is expected, since data and PDE loss conflict in the training objective, so that purely training on PDE yields a lower baseline. However, though this converges to a stable model, the L^2 error converges to roughly 14.653%. The no data model found a solution that respects the PDE, but does not match the data, suggesting that data itself is important for accurate predictions.

The best performing models were $N_x = 128, 64, 32$, which had a smaller relative L^2 error than the no-data baseline. As expected, the physics residual was higher, but did not explode. $N_x = 16$ is where performance starts to drop. The relative L^2 error climbs at higher resolutions, indicating that zero-shot super resolution starts to break down. For highly under-sampled training data ($N_x = 8, 4$), the data is so coarse that it seems counterproductive to the physics residual. This causes a lack of convergence and exploding errors at nearly all resolutions.

Ultimately, it seems that zero-shot super resolution generally holds until $N_x = 16$. Interestingly, this is where the number of modes is less than the capacity of our model, so we are not as expressive to carry to higher resolutions. Training data significantly beyond the number of modes does not seem to offer significant gain.

3 UTILITY OF THE DATA LOSS: BLACK-SCHOLES UNDER ILL-DEFINED REAL-WORLD DATA

Most existing applications of the PINO framework have been problems with high-fidelity data. Apart from aleatoric uncertainty from measurement tools or environmental stochasticity, system data fed as input for training conforms to the underlying solution function. However, such an assumption is often idealized, and requiring this discounts vast domains where data is messy. Fields in social sciences such as psychology, sociology, and economics often have problems loosely guided by physical models, but also incorporate human feedback. Humans are known to act irrationally, often governed by emotion rather than fact. As such, data in these domains do not necessarily agree with a mathematical law. We investigate whether the PINO architecture still provides value for such problems. In other words, whether sub-par real-world data could still aid in prediction.

We assessed this problem through the *Black-Scholes equations*. Coming from mathematical finance, the Black-Scholes equations provide the theoretical fair price of a European-style option. Given a fixed exercise price (S), volatility (σ), and time horizon (T), this partial differential equation relates the theoretical value of the option (V) as the stock price (S) changes:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0 \tag{1}$$

This equation has a known analytic solution 2. We first applied the PINO framework to solve the Black-Scholes equation with generated data from this solution 3.1. Then, we conducted a comprehensive assessment of performance of PINO under three different data domains: no data (as a baseline), messy real-world data, and analytical data 3.2. The goal is to better understand the role and value of training data in the PINO framework.

3.1 Solving Black-Scholes with an Analytical Data-generator

Derived using a transformation of variables to the classic 1D heat equation, the analytic solution of the Black-Scholes equation is given below:

$$C(S_t, t) = S_t N(d_1) - Ke^{-r(T-t)} N(d_2) \tag{2}$$

where $C(S_t, t)$ is the value of the call-option, S is the current price of the stock, K is the strike price, $T - t$ is the time to maturity, r is the risk-free interest rate, σ is the fixed volatility, and $N(x)$ is the cumulative distribution function (CDF) of the standard normal distribution.

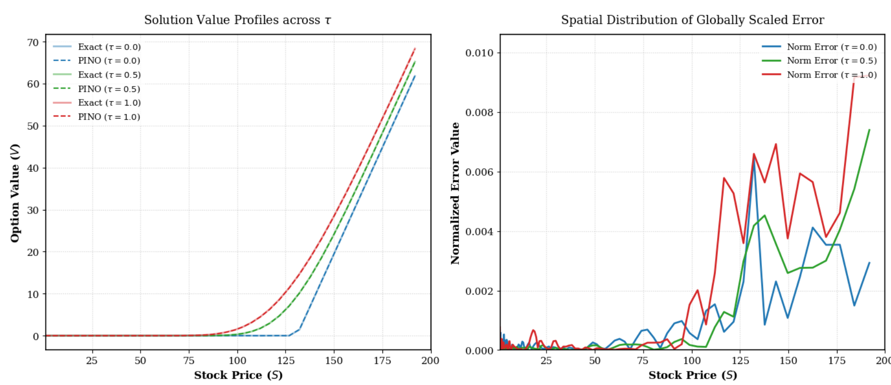


Figure 6: Comparing performance of the PINO framework to solve the Black-Scholes equation under varying normalized time-to-expiration τ.

We used this equation to generate 1000 equally-spaced true points along varying stock prices S. With this data, we trained PINO using 1 as the physics residual. This process was repeated for differing values of $\tau = \frac{1}{2}\sigma^2(T-t)$, which represents the normalized time-to-expiration. The results are shown in Figure 6. $\tau = 0$ corresponds to the expiration date and $\tau = 1$ corresponds to the initial purchase; as we interpolate from $\tau = 0$ to $\tau = 1$, the option value rises, corresponding to the time-value of money. For all three τ values, the predicted value closely matches the true solution, with a globally scaled error on order 1e-3. This implies that the PINO architecture successfully solves the Black-Scholes equation given sufficient, representative training data.

3.2 Integrating Real-world Data

Real-world market data is governed by humans, and often deviates drastically from the Black-Scholes model. In this section, we test the capabilities of predicting the value from Black-Scholes given real-world option pricing data. Option prices are sourced from Yahoo Finance using Apple Inc. (AAPL) call options. Specifically, we pull option deals from the most recent expiration period at the time of this paper. Since market data contains anomalies, we first filter the data to ensure the training data given to PINO are reliable, liquid, and reflect actual market values rather than statistical noise. We selected options that were near-the-money (S/K between 0.85 to 1.15), with a minimum 20 number of trades, a minimum price above \$0.10, and stock price between 100-600 for our chosen discretization grid. The data is split 70-30 to train and validation sets, stratified by expiration data so that both sets have data across the time horizon.

After filtering and splitting, the number of training data points was $N = 32$. To accommodate such sparse data, we used sparse supervision with an observation mask. Each unique strike price K in our dataset holds a tensor containing initial conditions over the entire grid, and a sparse vector holding option prices y_s in the grid. If a real option contract exists at strike K at specific expiration τ, the closest coordinate of our grid is mapped to a value in y_s. We have a mask which only sets values to 1, so that the model is only penalized for incorrectly computing data points for which a real market quote actually exists.

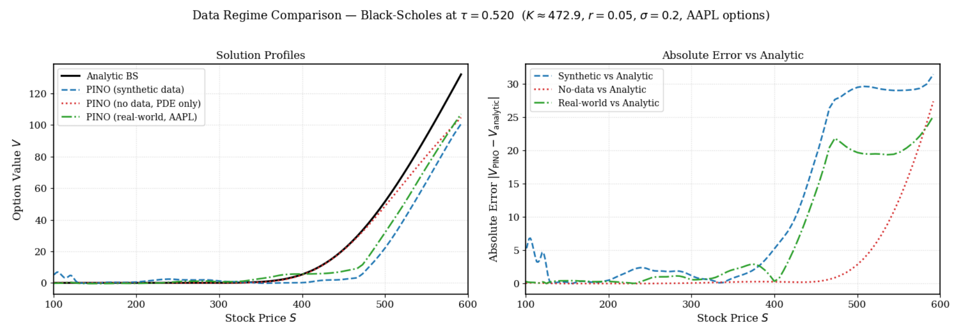


Figure 7: Performance of different PINO frameworks when solving the Black-Scholes equation under various data-regimes

Figure 7 compares performance of the solution profiles among three different data regimes: PINO with synthetic data, PINO with real-world AAPL data, and PINO with no-data (only physics loss). To create an apples-to-apples comparison, the synthetic data only consisted of $N = 32$ points. PINO with only physics loss closely matched the value up to the exercise price where it then deviates. Both the synthetic and real-world data models start sloping later than the analytic solution but seem to align with it, suggesting that both the synthetic and real-world data were sufficient to capture the trend. More research is likely needed, but these results suggest that messy data has the potential to align PINO models nearly as well as high-fidelity data.

4 Related Work

Prior work assessing zero-shot super resolution capabilities often explicitly trains the architecture to perform super-resolution through low-resolution and high-resolution (LR-HR) data pairs (Sinha et al., 2025; Wei & Zhang, 2023). Zero-shot super resolution is then measured on finer input grids. Another common line of work aims to test zero-shot super resolution without training data entirely, relying on physics constraints alone to describe the governing system (Kaewnuratchadasorn et al., 2024). Super-resolution is commonly assessed through multi-resolution, designing new custom architectures to accurately propagate physical information compared to standard interpolation (Kaewnuratchadasorn et al., 2024; Wei & Zhang, 2023). In many scenarios, the resolution at inference time is orders of magnitude higher than physics or data loss at training time (Sakarvadia et al., 2026). Our approach here does not embed high-resolution images during training, but instead assesses the capabilities of the model trained with identical input-output discretizations. We also never test a resolution that has not been seen during training, particularly using sampled PDE loss at or above the inference grid.

There has been many works applying the PINO architecture to tackle novel problems (Ding et al., 2026; Lydon et al., 2026; Cheng et al., 2025). However, work has primarily been in science and engineering domains. The data given here may be sparse due to fundamental limitations, but is expected to match closely with the underlying partial differential equation. In contrast, this work applies the framework to an economic domain where data is not uniform across the domain, and has low-fidelity to the Black-Scholes equation. We didn't know beforehand whether such data would help or hinder training, which formed the basis of our experiment.

Additionally, much work in this field focuses on fundamental changes to the PINO architecture itself, integrating recent developments in deep learning and numerical methods (Ganeshram et al., 2025; Wang et al., 2025; Ma & Alkhalifah, 2026; Boya & Subramani, 2025). Rather than transforming the structure of these networks, our work aims simply to stress-test the general framework under non-ideal conditions.

5 DISCUSSION

The Physics-Informed Neural Operator (PINO) architecture is supported by two fundamental contributions: data and physics loss terms. We found that the physics loss term is fairly robust, and helps the model propagate physical information from low coarse data across the entire domain. Even without a finer physics loss term, the model's error often converges uni-directionally as the resolution scales. Zero-shot super resolution appears to be robust, both in terms of the resolution magnitude it extends to and how low-coarse training data it requires to do so. Even at 32x the training data resolution, we found no dip in performance of the model as long as physics loss was sampled at that resolution. Moreover, the training resolution itself does not need to be fine, just as long as it does not obscure the underlying Fourier wavenumbers, which in practice is fairly low. The original PINO paper used $k = 12$ modes to get convincing results for complicated, coupled PDEs such as Burger's equation and Kolmogorov flows. While the paper only tested zero-shot super resolution on coupled Kolmogorov flows, we found that zero-shot super resolution works even when modes are decoupled, as in the 1D Wave Equation.

When controlling the number of training data points, we found that the PINO model trained with low-fidelity, real-world option prices did not significantly outperform the model whose data was generated from the analytic solution. Both learned solution functions had comparable error rates across the mesh, but fared better than a baseline model trained with no data. These findings highlight the importance of data in the PINO pipeline, suggesting that even sparse, low-fidelity data can provide a critical anchor for the solution profile that physics loss alone cannot replicate. Future work is needed to provide a rigorous framework for when data is too rough to train accurately.

6 CONCLUSION

In this paper, we aimed to determine how well the Physics-Informed Neural Operator holds up when each of its core components, data and physics loss, are degraded. For physics loss, we systematically scaled up the physics resolution from the initial training data, stressing the residual term to guide the solution function along finer meshes. We demonstrate that zero-shot super resolution is robust up to massive grid scale-factors, provided that the network's internal frequency representations are bounded by the informational content of the source data. For data, we highlight that models trained with real-world market data does not deviate significantly in accuracy compared to a model trained with ideal data sampled from the Black-Scholes analytic solution.

These results provide strong backing for the use of the PINO architecture in various domains, specifically in low data regimes where data may be messy or ill-defined. As pre-existing applications have mostly focused on problems with a known analytical solution, or systems whose data closely matches the PDE, this greatly expands the realm of applications. Domains in which collecting training is time-consuming, costly, or dangerous can have strong potential for this architecture. Furthermore, social sciences where real-world data may be confounded with unknown variables may still find value using this architecture for prediction.

The work done here is mainly empirical, on simple one-dimensional problems. Further work may be done to run these experiments on more complicated PDE equations. In particular, an analysis of which properties in a partial differential equation lend themselves to long-range super resolution in low data regimes or value through noisy real-world data would be incredibly valuable for this field. We suspect that some notion of how continuous or regular the solution function is may play a role, as well as its dimensionality. Nonetheless, we believe that these results offer promise to solve novel, equation-driven problems in a wide range of domains.

Author Contributions

Hisham conducted the experiments with zero-shot super resolution, including creating the code, generating the figures, and writing the relevant sections. He also polished the code for the Black-Scholes code and wrote most of that section. Also wrote the Introduction, Related Work, and Conclusion in this paper.

Muhammadbager conducted the experiments on the Black-Scholes application, including implementing the Crank-Nicolson finite difference solver for generating synthetic training data, building the pipeline to integrate sparse real-world options data from Yahoo Finance, and developing the masked sparse-data loss for training PINO on empirical market prices. Also generated the relevant figures and wrote the relevant sections.

Graham worked on implementing losses and fixing the full workflow for solving Black-Scholes with PINO. He also wrote the abstract and contributed to initial resolution experiments and applications of PINO, specifically related to shallow water equations.

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